Original Investigation

Comparison of Conventional, Revascularized, and Bioengineered Methods of Recurrent Laryngeal Nerve Reconstruction

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IMPORTANCE Damage to the recurrent laryngeal nerve (RLN) is highly detrimental to voice, swallow, and cough. The optimal method for reconstitution of a nerve gap after injury is unknown.

OBJECTIVE To evaluate multiple methods of RLN reconstruction.

DESIGN, SETTING, AND PARTICIPANTS This study used an established canine model of RLN injury to examine purpose-bred, conditioned, female, 20-kg mongrel hounds at Washington University. A total of 32 dogs were examined, with 63 experiments performed.

INTERVENTIONS Surgical transection or excision of the RLN with reconstruction by multiple methods.

MAIN OUTCOMES AND MEASURES Six months after injury repair, laryngeal adductor pressures (LAPs), spontaneous and stimulable movement, and graft axon counts by histologic analysis were assessed.

RESULTS Simple RLN transection with direct neurorrhaphy provided a mean (SD) recovery of 55.5% (12.5%) of baseline LAPs (P = .18 for comparison of LAP recovery among cases from the conventional nerve graft [39.4% (22.2%)]; P = .63 for comparison of LAP recovery among cases from the reverse autograft [60.8% (27.5%)]). Revascularized grafts provided a recovery of 54.5% (46.4%) while short and long acellular grafts provided recoveries of 60.4% (NA) and 39.5% (17.0%). Two of 11 polyglycolic acid reconstructions provided a measurable LAP with a mean (SD) recovery of 37.1% (8.9%) of baseline. Reconstruction with a neural conduit in any condition provided no measurable LAP recovery.

CONCLUSIONS AND RELEVANCE Conventional nerve grafting resulted in no significant difference in recovery of LAP function compared with simple neurorrhaphy or reverse autograft. Conventional and revascularized nerve grafts provided similar recovery. The use of bioengineered acellular nerve grafts or nerve conduits for reconstruction resulted in poor recovery of function.

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JAMA Otolaryngol Head Neck Surg. 2016;142(6):526-532. doi:10.1001/jamaoto.2016.0151
Published online May 5, 2016.

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Although there is some inherent capability for nerves to regenerate, recovery after damage to a peripheral nerve is frequently unpredictable and can result in a poor outcome. Often, when a nerve is completely transected, the distal and proximal stump can be directly repaired via microsurgical suturing. However, in many cases, primary nerve repair may not be feasible, such as after a trauma or resection during a cancer extirpation. In these situations, a bridging component of tissue must be used to provide a tension-free anastomosis.[1] When selecting a graft, surgeons will typically choose to use an autologous nerve graft harvested from another site on the patient's body. The use of an autograft, however, is limited by size mismatch among the nerve tissues and lack of sufficient donor sites.[2,3] However, several of these types of grafts exist, including skeletonized cable grafts, acellular nerve graft substitutes, or vascularized nerve grafts. When these nerves are repaired under tension, there is almost always a poor outcome.[4,5]

Numerous other factors influence the ultimate success of nerve grafting, including the diameter of the nerve graft.[6] Nerve grafts have central necrosis that is thought to be detrimental to the regeneration of the advancing axons.[6] One currently accepted technique for avoiding central necrosis would be to place the nerve in a well-vascularized bed.[6] This technique assumes that the vascularized bed is sufficient to sustain the graft and ultimately cause minimal damage. Within the head and neck, most sites are closely related to the large vessels of the neck, but sites of relatively low blood flow exist, particularly around the trachea. Another proposed technique involves the use of vascularized nerve grafts that contain the surrounding vessels in addition to the neural tissue. These vessels are anastomosed to the surrounding vasculature to help provide native blood flow to the graft. A previous study[6] found that most grafts will obtain blood flow after 72 hours within the recipient site but that there is substantial blood flow even earlier in vascularized nerve grafts, suggesting there may be better performance of the graft. Within the upper extremities, substantial work has indicated that patients who received vascularized grafts do better than patients with nonvascularized grafts in scarred recipient beds or for longer nerve gaps.[7]

The recurrent laryngeal nerve (RLN) is sometimes sacrificed while the patient is undergoing thyroidectomy for a malignant tumor or during thoracic procedures. This nerve is responsible for the movement of the ipsilateral glottis, and its loss can be highly detrimental to a patient's voice, swallow, and cough. When a nerve is transected, the ideal reconstruction is with direct neurorrhaphy as seen in previous study.[8] However, when a gap exists in the nerve, this will frequently require grafting.[9] Often, sacrifice of the nerve is unplanned, and as such, repair of the nerve will need to be immediate to ensure the lowest amount of morbidity associated with its loss. Techniques typically used by head and neck surgeons include using free ansa nerve cable grafts or great auricular nerve cable grafts.[10] One group reported the use of free ansa grafts followed by vein wrapping and ultimately found better results than with conventional reconstruction.[11] An additional study[12] found that immediate reconstruction of the RLN during surgery using grafts or direct anastomosis produced reasonable voice outcomes. Although these methods function to a certain extent, they can certainly be improved by evaluating other known methods of nerve repair used in other situations.

Key Points

Question What is the comparative efficacy of repairing a gap in the recurrent laryngeal nerve (RLN) with a reverse autograft, conventional autograft, revascularized autograft, acellular allograft, or bioengineered conduit?

Findings In this canine, surgical intervention study, there were no significant differences in recovery of laryngeal function for conventional or revascularized RLN grafting compared with neurorrhaphy or reverse autograft. The use of acellular nerve grafts or nerve conduits for RLN reconstruction resulted in poor recovery of laryngeal function.

Meaning For repairing recurrent laryngeal nerve gaps, use of a conventional nerve graft provides similar recovery to that of a revascularized graft, whereas acellular nerve grafts or nerve conduits provide an inconsistent functional recovery.

Previous studies[13,14] have also found that short (1- to 2-cm) motor nerve gaps can be repaired by the use of empty polyglycolic acid conduits through which regenerating axons find their distal targets. Silicone tubes loaded with nerve growth factors in a heparin affinity–based fibrin delivery system have also enhanced motor nerve recovery across critical gaps in rat models.[15,16] Acellular nerve grafts have been used for both short and long gaps and are now available commercially.[17,18] Revascularized nerve grafts are also advantageous in repairing long nerve gaps.[19,20] In this study, we tested all of these methods for reconstruction of the RLN, comparing them with standard nonvascularized autografts.

Methods

This study used an established canine model of RLN injury to examine purpose-bred, conditioned, female, 20-kg mongrel hounds.[21] A total of 32 dogs were examined, with 63 experiments performed. Each of the animals in the study received standard care within US Department of Agriculture, National Institutes of Health, and Office of Laboratory Animal Welfare animal treatment guidelines verified by the Animal Studies Committee and the Office of Animal Affairs at Washington University. All experiments were performed according to a protocol approved by the Institutional Animal Care and Use Committee.

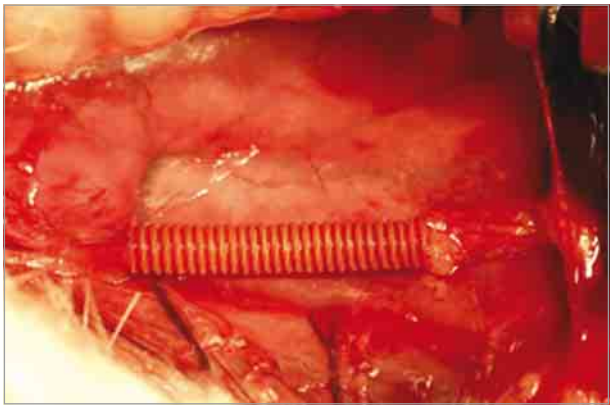
Surgery

With the animal under general anesthesia, the neck was opened using a midline incision, and a permanent tracheostomy was performed according to our previous method through tracheal rings 10 to 13.[22] After this procedure, both RLNs were dissected out, an electrode was placed on 1 RLN, and another endotracheal tube was inserted between the vocal folds to measure laryngeal adductor pressures (LAPs) as the RLN was stimulated at frequencies ranging from 20 to 100 Hz, as previously described.[23] The RLN was stimulated for 3 to 5 seconds at a series of fixed-interval

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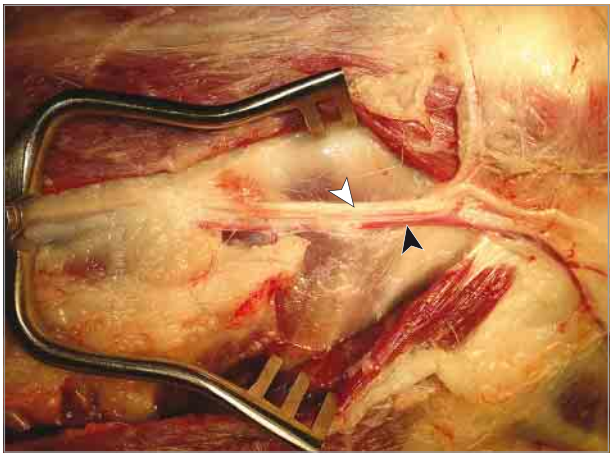
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Figure 1. Polyglycolic Acid Tube In Vivo



A total of 0.5 cm of each of 2 recurrent laryngeal nerve stumps is sutured into a 2-cm segment of a polyglycolic acid tube after 1 cm of the nerve segment has been removed. A 1-cm gap exists between the 2 nerves inside the polyglycolic acid tube.

Figure 2. Saphenous Neurovascular Bundle In Vivo



A dissected canine hind limb with saphenous nerve (white arrowhead) and its associated vasculature (black arrowhead).

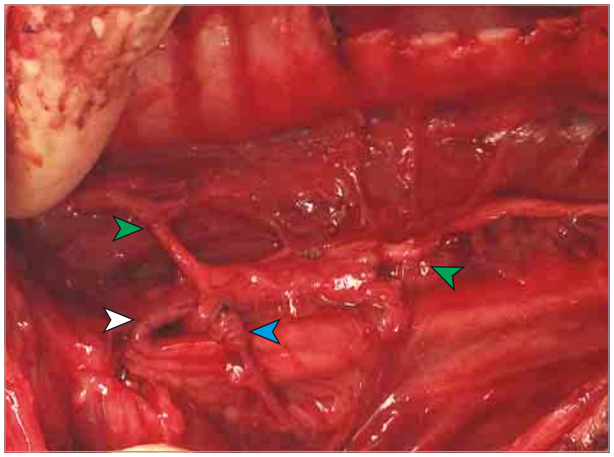
stimulation levels using a customized constant-current laryngeal nerve stimulator (WR Medical Electronics). The nerve was allowed to recover for at least 30 seconds between stimulations. The values measured during this initial procedure provided the pretreatment baseline LAP curve. At 6 months after injury, the canines were anesthetized and the process performed again to obtain postinjury LAPs.

After initial measurement of the LAPs, the RLNs were injured and repaired via a variety of randomly selected methods. All microneural surgery was performed under an operating microscope, and all anastomotic sutures used were 9-0 nylon.

Controls

The RLN was transected and immediately reanastomosed with interrupted sutures (8 dogs, 16 total experiments) as previously described.[8]

Figure 3. Revascularized Nerve Graft In Vivo



Dissected recurrent laryngeal nerve after removal of 5 cm of nerve and placement of a vascularized saphenous nerve graft showing 2 nerve anastomoses (green arrowheads), arterial anastomosis (white arrowhead), and vein anastomosis (blue arrowhead).

Short Gap Experiments

For each study group (n = 7), the RLN was transected twice, removing a 1-cm segment. The nerve endings were then sutured to 2-cm polyglycolic acid tube (Neurotube, Synovis Life Technologies) nerve conduits. The 2-cm nerve conduits had 0.5 cm of the transected nerve endings loaded into each side, leaving a 1-cm gap (**Figure 1**). The rest of the polyglycolic acid tube was then filled with normal saline solution (n = 11).

The nerve endings were sutured into a 2-cm-long, 1.5-mm-diameter polymeric silicone tubing, leaving a 1-cm gap. One conduit was filled with saline (n = 4) and another with bioengineered carrier only (n = 4). The carrier was prepared from a previously published protocol, which includes fibrin matrix and a heparin affinity–based delivery system.[24,25] The conduits filled with the bioengineered carrier were loaded with a glial cell–derived neurotrophic factor (n = 8) or neurotrophin 3 (n = 8).

The nerve endings were sutured to a 1-cm-long acellular nerve graft (n = 4). Acellular nerve allografts were harvested from canine saphenous nerves, which were chemically engineered via the protocol of Hudson et al.[26,27]

For the reverse autograft, the excised nerve segment was rotated 180° and sutured back into place (n = 8).

Long Gap Experiments

For each of these study groups, the RLN was transected twice, removing a 5-cm segment. All animals (n = 8) received a weight-based dose (0.5 mg/kg) of dexamethasone at the beginning of their initial procedure. A 5-cm segment of saphenous nerve was then harvested and sutured at each end (n = 6). A 5-cm graft that included the saphenous nerve, artery, and vein was then harvested. Microvascular anastomoses were performed end to end to the cranial thyroid artery and internal jugular vein (**Figure 2** and **Figure 3**). These animals received 81 mg of aspirin from postoperative day 0 until postoperative day 5 (n = 4). Acellular nerve allografts were prepared as described above and sutured at each end (n = 6).

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Follow-up

At 1 to 4 weeks postoperatively, the vocal fold movement was examined with an infraglottic examination in the awake animal by using a rigid telescope inserted through the tracheostomy site and directed superiorly in the awake dog. Complete vocal fold paralysis on the treated side was confirmed in all cases.

At 6 months after the operation, infraglottic examination was performed again to record spontaneous vocal-fold movement. With the animal under general anesthesia, the neck was explored inferior to the tracheostomy site, and the RLNs were dissected (at a previously undissected inferior level). Electrodes were placed distally, and the vocal folds were observed via direct laryngoscopy. Movement with a range of stimulation parameters was recorded via direct laryngoscopy, and the LAP curves were recorded as performed previously. The RLNs were then dissected distally to inspect the site of the reconstruction. The movement was scored on a grade of 0 to 5, with 0 indicating no twitch; 1, brief twitch; 2, slight adduction; 3, adducts but does not reach midline; 4, adducts to midline; and 5, adducts and abducts completely.

Histologic Analysis

Histologic analysis was performed via the Washington University School of Medicine histology core. After the nerves (both control and experimental) were removed from the canines, the nerves were immersed in 4% paraformaldehyde fixative. Grafts and normal nerve specimen were subsequently embedded in Epon medium, placed on slides in 40-µm sections, stained with osmium tetroxide, and counterstained with hematoxylin-eosin. The slides were then microscopically evaluated and individual axons counted to quantitatively evaluate axonal regeneration (**Figure 4**).

Statistical Analysis

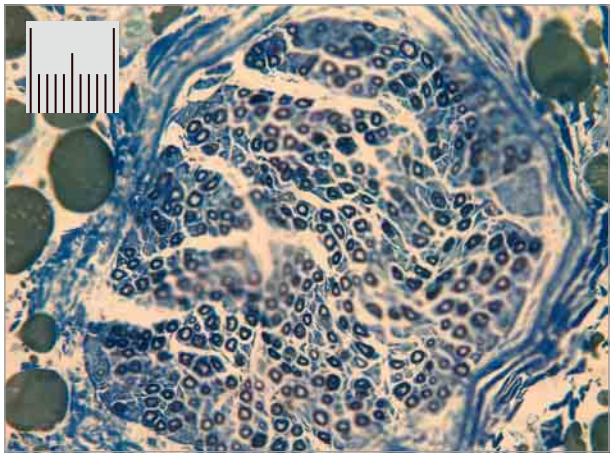
Laryngeal adductor pressure values were calculated by computing the mean initial laryngeal adductor pressure (in mm Hg) and the mean final recovered pressure (in mm Hg) when the recurrent laryngeal nerve was stimulated at each of the following frequencies: 70, 80, 90 and 100 Hz. The mean baseline and mean final laryngeal adductor pressure values were then compared for each individual experiment, providing a percentage recovery from baseline. A mean and standard deviation were calculated for each group utilizing the percentage recovery of laryngeal adductor pressure for the individual experiments in that group. A student's *t* test was performed to derive a *P* value for each comparison group.

Results

All of the dogs (n = 32) survived the treatment time. Each of the dogs receiving vascular nerve grafts developed wound infections at their leg donor sites; therefore, this experiment was stopped early at 4 experiments of a planned 6. One dog developed stomal stenosis and required a stomal revision.

Data are summarized in the **Table**. The LAP data are presented as 2 averages: 1 for the successful repairs only (defined as having a measurable LAP value) and 1 averaging all of the values, including the unsuccessful repairs. The control comparison group, comprising simple RLN transection with direct neurorrhaphy, provided a mean (SD) recovery of 55.5% (12.5%) of baseline LAPs, which has previously been reported.[8] The repair of short nerve gaps (1-2 cm) provided a variety of results. Two of 11 polyglycolic acid tube reconstructions provided a measurable LAP with a mean (SD) recovery of 37.1% (8.9%) of baseline for successful repairs or 6.7% (15.3%) for all repairs. Reconstruction with a polymeric silicone neural conduit in any condition provided no measurable LAP recovery. Short acellular nerve allografts provided a measurable LAP in 1 of 4 reconstructions, with a mean recovery of 60.4% of baseline or a mean (SD) recovery of 15.1% (30.2%) when unsuccessful repairs are included. Reverse autografts provided a mean (SD) recovery of 60.8% (27.5%) of the baseline LAPs.

Figure 4. Axon Histologic Analysis



Postinjury recurrent laryngeal nerve showing staining of the axons for counting (osmium tetroxide and hematoxylin-eosin, original magnification ×40). Scale bar indicates 1 mm.

Conventional grafting with a 5-cm nerve graft was successful in 5 of 6 repairs and provided a mean (SD) LAP recovery of 32.8% (25.6%) of baseline in all repairs or 39.4% (22.2%) when averaging the successful repairs. Revascularized nerve grafting of 5 cm was successful in 3 of 4 experiments and provided a mean (SD) LAP recovery of 40.9% (46.7%) in all repairs or 54.5 (46.4%) in successful repairs only. Acellular nerve grafting of 5 cm was successful in 2 of 6 repairs and provided a mean (SD) LAP recovery of 13.2% (21.8%) in all repairs or 39.5% (17.0%) in successful repairs. All revascularized nerve grafts were noted to have a patent blood supply at canine death.

When comparing the LAP recovery in all cases of 5-cm conventional grafting with the direct neurorrhaphy group or the reverse autograft groups, there was no statistically significant difference (32.8% vs 55.5%, $P = .08$; or 32.8% vs 60.8%, $P = .07$, respectively). Use of the 2-cm reverse autograft appears to provide an equivalent repair to direct neurorrhaphy (60.8% vs 55.5%, $P = .51$). Reverse autografts provided a better repair than the polyglycolic acid conduit (55.5% vs 6.7%, $P < .001$) or an acellular graft (55.5% vs 15.1%, $P = .04$). No statistically significant differences could be identified when

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Table. Methods of Experimental Nerve Repair

Recurrent Laryngeal Nerve Repair Method	No. of Experiments	No. With Stimulable Movement	No. of Axons per Graft, Mean (SD)	Mean Awake Laryngeal Movement (Scale, 1-5)	Mean Stimulated Laryngeal Movement (Scale, 1-5)	No. (%) of LAPs of Successful Cases, Mean (SD)	*P* Value	No. (%) of LAPs of All Cases, Mean (SD)	*P* Value
Transection with direct neurorrhaphy[a]	16	16	NA	2.6	3.7	55.5 (12.5)	.18,[b] .63[c]	55.5 (12.5)	.08,[b] .63[c]
Short (1- to 2-cm) gap repair									
Polyglycolic acid conduit	11	2	NA	1.0	3.3	37.1 (8.9)	.09[c]	6.7 (15.3)	<.001[c]
Polymeric silicone conduit with empty carrier	4	0	NA	0	0	0	NA[c]	0	NA[c]
Polymeric silicone conduit with fibrin-embedded carrier	4	0	NA	0	0	0	NA[c]	0	NA[c]
Polymeric silicone conduit with fibrin-embedded GDNF carrier	8	0	NA	0	0	0	NA[c]	0	NA[c]
Polymeric silicone conduit with fibrin-embedded neurotrophin 3 carrier	8	0	NA	0	0	0	NA[c]	0	NA[c]
Short acellular nerve allograft	4	1	NA	2.0	5.0	60.4 (NA)	NA[c]	15.1 (30.2)	.04[c]
Reverse autograft[c]	8	8	NA	2.6	3.5	60.8 (27.5)	.16[b]	60.8 (27.5)	.07[b]
Long (5-cm) gap repair			NA						
Conventional cable graft[b]	6	5	182 (66)	2.5	2.6	39.4 (22.2)	.16[c]	32.8 (25.6)	.07[c]
Revascularized nerve graft	4	3	202 (75)	2.8	3.5	54.5 (46.4)	.69[b]	40.9 (46.7)	.77[b]
Acellular (bioengineered) nerve allograft	6	2	0	1.3	1.2	39.5 (17.0)	.99[b]	13.2 (21.8)	.18[b]

Abbreviations: GDNF, glial cell–derived neurotrophic factor; LAP, laryngeal adductor pressure; NA, not applicable.

[a] Data are from a prior study.[8]

[b] *P* values from a *t* test are given for comparison of LAP recovery among cases from the conventional nerve graft.

[c] *P* values from a *t* test are given for comparison of LAP recovery among cases from the reverse autograft.

comparing the conventional cable graft repair with revascularized grafting or acellular nerve grafting.

With regard to the histologic data, the mean for all measured 5-cm grafts revealed that approximately 66.5% (range, 42.1%-84.5%) of axons traverse the grafts. The conventional nerve grafts had a mean (SD) of 182 (66) axons in the graft segment. Revascularized grafts had a mean (SD) of 202 (75) axons in the graft segment, revealing no statistical significance between the groups (*P* = .77). The mean number of axons in an unaffected canine RLN is approximately 285. For the acellular grafts, no axons could be identified in the nerve graft portion on histologic analysis even though 2 of these grafts had a positive, although weak, LAP. No statistically significant difference was apparent between the vascular and avascular nerve grafts with regard to their histologic axonal appearance on microscopic evaluation.

Discussion

The use of vascularized nerve grafts, acellular nerve grafts, neural conduits, or nerve growth factors in the reconstruction of cranial nerves has been extremely limited and has only been described in case reports or investigational studies.[19,20,28] This study looked at using these advanced nerve repair techniques for short and long gaps of the RLN. The standard repair for a nerve gap includes the use of an autologous nerve graft. However, a number of inherent disadvantages exist to using autologous nerve grafts, including the limited availability of a suitable nerve and donor site morbidity, such as scarring and numbness.[17,18] The US Food and Drug Administration–approved and commercially available acellular grafts and conduits are currently used for multiple indications. Acellular nerve grafts have great promise for peripheral nerve repair in that they are easy to obtain (available ready to use in frozen sterile package), avoid donor morbidity, and have demonstrated equivalent results in some cases. In the current study, we found that these grafts do not work well in the repair of the RLN for long or short gaps. In the repair of larger nerve gaps, the acellular graft only functioned 33% of the time, overall providing a very weak recovery of function in the RLN. The repair of short gaps was similarly low, with only 25% of canines recovering some function. Ultimately, our study found that these grafts did not provide a consistent recovery of function for RLN defects.

Other grafting techniques for short nerve gaps, including the use of polymeric silicone or polyglycolic acid conduits, have also been proposed with some success in treating peripheral nerve injuries.[29] Further loading of these conduits with nerve growth factors, including glial cell-derived neurotrophic factor and neurotrophin 3, has been attempted with improved results in nerve regeneration.[24,30,31] Of interest, the coupling techniques used in this study did not provide any consistent or significant recovery of nerve function for short nerve gaps. Similar to acellular nerve grafts, many

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of the previous studies[24,30,31] reported the recovery of peripheral nerves with use of these conduits, but cranial nerve recovery has not been thoroughly investigated when using these devices.

Grafting for long nerve grafts revealed that conventional cable grafting had a similar outcome to the revascularized nerve graft. There was essentially no evident benefit to providing additional blood flow to the nerve graft. Axon counts were also similar for these 2 differing reconstructions. It is unclear why both 1 vascular and 1 conventional nerve graft failed to generate a LAP in the study when reconstructing a long nerve gap. Each of these grafts was surgically dissected post mortem and noted to have intact nerve anastomoses. In addition, for the RLNs reconstructed with acellular nerve grafts, it is unknown why no axons could be identified on pathologic analysis of these nerves while there was an identifiable LAP.

Some of the weaknesses of this study include low animal numbers in each group, which precludes strong statistical analysis. However, the surgical techniques and outcomes for these large animals more closely approximate the human condition than lower vertebrate studies. Therefore, even though the treatment numbers are low, the results are more likely to be able to translate into human therapeutics. Another weakness is that the data provided a wide range of results, with many of the experiments revealing no viable recovery of the LAPs. However, there are trends identifiable even if statistical significance cannot be reached.

This is the first study, to our knowledge, to compare the use of vascularized nerve grafts, acellular nerve grafts, and specialized coupling techniques with nerve growth factors for RLN repair for short and long nerve gaps. This study revealed that use of polyglycolic acid conduits (2 of 11 successful) or polymeric silicone (0 of 24 successful) for the repair of short gaps in the RLN did not provide a consistent or reliable result. The use of the short acellular nerve graft (1 of 4 successful) provided a limited response, but the result was also not consistent among our experiments. Vascularized nerve grafts did not appear to provide any improved benefit for long nerve graft reconstruction when compared with simple cable grafting. Moreover, acellular nerve grafts for long gaps provided poor and inconsistent recovery as well.

Conclusions

Conventional nerve grafting did not provide significantly different recovery of LAPs after RLN repair when compared with a simple neurorrhaphy or a reverse autograft. Revascularized nerve grafts provided similar recovery benefit for long nerve gap reconstruction when compared with conventional grafting. The use of decellularized nerve grafts for reconstruction of long or short nerve gaps and nerve conduits for reconstruction of short nerve gaps all resulted in poor and inconsistent recovery of RLN function.

ARTICLE INFORMATION

Accepted for Publication: January 16, 2016.

Published Online: May 5, 2016. doi:10.1001/jamaoto.2016.0151.

Author Contributions: Drs Sand and Paniello had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.
Study concept and design: Sand, Park, Desai, Sakiyama-Elbert, Paniello.
Acquisition, analysis, or interpretation of data: Sand, Park, Bhatt, Desai, Marquardt, Paniello.
Drafting of the manuscript: Sand, Bhatt, Desai.
Critical revision of the manuscript for important intellectual content: All authors.
Statistical analysis: Sand.
Obtained funding: Sakiyama-Elbert, Paniello.
Administrative, technical, or material support: Sand, Park, Bhatt, Desai, Marquardt, Paniello.
Supervision: Park, Desai, Paniello.

Conflict of Interest Disclosures: All authors have completed and submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Dr Sakiyama-Elbert reported that she may receive royalty income based on a heparin affinity–binding delivery system (fibrin gels) that she developed, which was licensed by Kuros Therapeutics. That technology is evaluated in this research. No other disclosures were reported.

Funding/Support: This study was supported by grant R01 DC010884 from the National Institutes of Health (Dr Paniello).

Role of the Funder/Sponsor: The funding source had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Previous Presentation: This study was presented at the Annual Meeting of the American Head and Neck Society; April 26, 2015; Boston, Massachusetts.

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